

	Arm 1 – Software mathematical audit without access to raw data or identifiers (hash-only)	Arm 2A – Standard operating protocol (SOP) audit	Arm 2B – Proportional response software + SOP augmented audit
What you receive	Exported hash-only packets ≈ 3.3MB per corpus; ≈ 800 files no raw files and no identifiers	Complete frozen corpus per tier ≈ 33.0MB per corpus; ≈ 254 files PDFs + CSVs + JSON + audit trail	Full corpus per tier PLUS AuditLog.AI and QMSAuditor_v5 Professional software
What you do	Verify exported AuditLog.AI hash-only packets using QMSAuditor_v5 Lite	Apply your SOPs with no restrictions	Software enumeration + SOPs (guided proportional review)
Corpora	Tk + 2 negative controls + T1 + T2 + T3 Labelled Tk, blinded QMS_01-05	As per Arm 1; each arm is independently randomized (blind)	As per Arm 2
Reporting	Timed analyses; MATCH/MISMATCH ; divergence enumeration files + memberships	As per Arm 1	As per Arm 1
Data class	Hash-packets publicly available via Mendeley (no raw data or identifiers)	<u>Simulated</u> trial data restricted to participating institutions	As per Arm 2
Ethics	Not required. Simulated trial / synthetic subjects (no real patients)	As per Arm 1	As per Arm 1
Privacy	No raw data or identifiers transmitted by software	Outcomes disclosure gated by participating institutions	As per Arm 2
Time	Software near instantaneous; Human variable. Estimated ~0-10min/corpus	Variable, as per SOPs (estimated ~1–4 hrs/corpus)	Proportional review hypothesis (guided by software): Arm2B<2A
Recognition	Principal investigator of participating institutions will be recognized as co-investigators		

Arm 1 – Software mathematical audit without access to raw data or identifiers: Participating institutions receive zero-custody hash-only packets: no raw files, no filenames, no clinical content (cryptographic fingerprints only: SHA-256 / RIPEMD-160 / Evidence Set Fingerprints). Operators use QMSAuditor_v5 Lite (open access, offline, no data collection; <https://auditlog.ai/qms-auditor-lite>) to run paired comparisons against the pre-anchored reference state; then use software to time human verification effort required to confirm correctness of software outputs. Verification outcome is a binary MATCH (proof-of-unchanged) or MISMATCH (divergences enumerated at the file and folder-membership layers).

Arm 2A – Standard operating protocol (SOP) audit: Participating institutions receive the complete frozen (pre-committed) corpora for each evidence-state: PDFs, CSVs, JSON, audit trail. No specific auditing methodology is mandated (unrestricted, as per institution's own standard operating protocols). Teams use their QA workflow; any tools they choose other than those provided by AuditLog.AI (reserved for Arm 1/2B). The only procedural requirement is deterministic time recording via the unlock-event commit and audit-report commit chain. This design maximises ecological validity; the goal is to characterise real-world audit performance, not a controlled method variant.

Arm 2B – Proportional response, software + SOP augmented audit (optional exploratory sub-study): Arm 2B is offered to any Arm 2A site that can allocate an additional independent auditor (an individual who has not participated in or been debriefed on the site's Arm 2A audit). Following submission and lock of the Arm 2A verdict, the Arm 2B auditor receives: (1) the same complete frozen corpora distributed to Arm 2A; and (2) unrestricted access to anchoring software AuditLog.AI and verification software QMSAuditor_v5 Professional (full version, Modes B and C enabled). The auditor may then anchor each corpus, export hash-packets and run paired comparisons. The hypothesis is that deterministic divergence enumeration augments SOP by guiding proportional response, reducing human time while maintaining or increasing accuracy. Software can be run for unlimited sessions during the evaluation window (up to 8-weeks), participating institutions are additionally encouraged to test the system using their own tests (tamper introduction) and controls (datasets).

Eligibility: Participation is restricted to Contract Research Organizations (CROs), and accredited academic or research institutions. Participants must have custodial responsibility for clinical, regulated, inspectable, or compliance-relevant evidence.

Duration: Each participating institution operates within a 4-week evaluation window from the date materials are distributed (Arm 1 and Arm 2A). A 4-week extension is available for Arm 2B on request, for a maximum of 8 weeks from activation.

Important (no PHI/PII transfer; no filename/path transfer required): Anchoring software operates locally; no source data or identifiers are transmitted. The only data transmitted is the institution's pseudonymous ID (e.g. QMSIND01), hash fingerprint and anchoring transaction ID for each completed session.